

NightRestore: A NAFNet-Based Universal Night Image Restoration System with Adaptive Multi-Degradation Pipeline

Prakhar Anil Sharma

PRN: 20250804051

20250804051@dypiu.ac.in

Atharva Jaind

PRN: 20250804011

20250804011@dypiu.ac.in

Avanti Kate

PRN: 20250804012

20250804012@dypiu.ac.in

Prajwal Deshpande

PRN: 20250804050

20250804050@dypiu.ac.in

Niraj Chavan

PRN: 20250804125

20250804125@dypiu.ac.in

Department of Computer Applications

D Y Patil International University, Pune

Abstract—Night-time and degraded image restoration is a challenging problem due to the simultaneous presence of multiple degradations such as low illumination, haze, noise, motion blur, lens flare, and rain streaks. Existing state-of-the-art methods either target a single degradation or require high computational cost. In this paper, we present NightRestore, a unified image restoration system built on the Nonlinear Activation Free Network (NAFNet) baseline proposed by Chen et al. [1]. Our system, called DirectNAFNet, extends the original NAFNet architecture with: (1) a multi-task training regime on heterogeneous datasets (LOL-v2 and Flare7K), (2) an automatic degradation detector based on luminance, contrast, and sharpness statistics, (3) a classical Dark Channel Prior defogging pre-processing stage, (4) a sliding window inference strategy with Hanning-weighted blending to handle arbitrary resolution inputs, and (5) an adaptive post-processing pipeline incorporating bilateral denoising, CLAHE, and conservative brightness scaling. Trained on the LOL-v2 paired low-light dataset (500 image pairs) for 30 fine-tuning epochs, our final model achieves a PSNR of 21.31 dB and SSIM of 0.823 on the LOL-v2 validation set, representing a significant improvement over intermediate ablation versions. The system is deployed as an interactive Gradio web application and hosted on Hugging Face Spaces. We compare our approach to the original NAFNet baseline and show that the proposed additions improve visual quality without introducing additional model parameters beyond the base 10.6M parameter DirectNAFNet.

Index Terms—Image Restoration, Low-Light Enhancement, NAFNet, Night Vision, Defogging, Deep Learning, Multi-Degradation

I. INTRODUCTION

Images captured under night-time or adverse conditions suffer from a combination of degradations that severely limit

their usability for human viewing, downstream computer vision tasks such as object detection, and medical imaging applications. The major degradation types encountered in real-world low-light scenarios include:

- **Low illumination:** Overall dark appearance with suppressed detail visibility.
- **Gaussian / Poisson noise:** Random pixel fluctuations due to photon shot noise at low light levels.
- **Atmospheric haze / fog:** Scattering effects that reduce contrast and saturation.
- **Motion blur:** Camera shake or moving subjects resulting in blurry captures.
- **Lens flare:** Scattered light from bright sources causing streaks and halos.
- **Rain streaks:** Visible precipitation artifacts.

Traditional approaches address each degradation independently. For example, Retinex-based methods [7] decompose illumination from reflectance to enhance low-light images, while Dark Channel Prior (DCP) [6] is widely used for image dehazing. However, real night-time scenes rarely present a single clean degradation type; instead, multiple degradations co-exist and interact, making independent processing sub-optimal.

Recent deep learning methods such as MPRNet [3], HINet [4], Restormer [2], and NAFNet [1] have demonstrated state-of-the-art performance on individual restoration benchmarks. Of these, NAFNet is particularly appealing due to its simplicity: it achieves competitive performance while eliminating

all nonlinear activation functions, instead relying on element-wise multiplication (SimpleGate) and a Simplified Channel Attention (SCA) module.

In this work, we leverage NAFNet as our backbone and propose **NightRestore**, a system that extends it to handle the full spectrum of night-time degradations. Our contributions are as follows:

- 1) We adapt the NAFNet block into **DirectNAFNet**, a lightweight variant (10.6M parameters, width=32) trained end-to-end for night restoration.
- 2) We design a **NightSynthesizer** augmentation pipeline that generates diverse multi-degradation training pairs from clean images, including synthetic rain, fog, noise, and flare overlays.
- 3) We introduce an **automatic degradation detector** that classifies each input image at runtime and activates appropriate pre- and post-processing modules.
- 4) We integrate a classical **Dark Channel Prior defogging** stage as a pre-processing step for foggy inputs, decoupling haze removal from the neural network's learned restoration.
- 5) We employ a **Hanning-weighted sliding window inference** strategy for arbitrarily sized inputs, avoiding boundary artifacts.
- 6) We implement an **adaptive post-processing pipeline** including bilateral filtering, adaptive CLAHE, color cast correction, and conservative brightness scaling.
- 7) We perform an **ablation study** tracking PSNR across multiple training versions, demonstrating progressive improvement from 13.30 dB to 21.31 dB.

II. RELATED WORK

A. Image Restoration with Deep Learning

Deep learning methods for image restoration have evolved rapidly from single-task feedforward networks to complex multi-stage architectures. Early CNN-based methods like DnCNN [5] demonstrated strong denoising via residual learning. Multi-scale U-Net architectures became the dominant paradigm, adopted by HINet [4], MPRNet [3], and others for their ability to process features at multiple resolutions.

B. NAFNet: Nonlinear Activation Free Network

Chen et al. [1] proposed NAFNet as a minimal yet powerful baseline for image restoration. The key insight is that nonlinear activation functions (ReLU, GELU, Sigmoid) are not necessary for state-of-the-art performance. Instead:

- **SimpleGate** replaces GELU: the feature map is split along channels and the two halves are multiplied element-wise, introducing nonlinearity without an activation function.
- **Simplified Channel Attention (SCA)** replaces the full Channel Attention module: global average pooling is followed by a single 1×1 convolution, and the result is multiplied channel-wise with the feature map.

NAFNet achieves 40.30 dB PSNR on SIDD for denoising (surpassing Restormer by 0.28 dB) and 33.69 dB PSNR on

GoPro for deblurring (surpassing MPRNet-local by 0.38 dB), while being computationally efficient (65 GMACs vs 778 GMACs for MPRNet).

C. Low-Light Image Enhancement

Retinex-theory based methods decompose images into illumination and reflectance maps. RetinexNet [8] introduced a paired low-light dataset (LOL) and a learnable decomposition network. More recent work uses data-driven approaches on paired datasets such as LOL-v2, which contains 500 training pairs of real low-light and normal-light images.

D. Image Dehazing

The Dark Channel Prior [6] is a classical statistical prior observing that in haze-free outdoor images, at least one color channel has very low intensity in local patches. This prior enables estimation of the transmission map and atmospheric light, allowing haze removal. While deep learning has improved upon DCP in accuracy, DCP remains effective as a fast pre-processing step.

E. Flare Removal

Dai et al. introduced the Flare7K dataset [9] containing thousands of scattering and reflective flare patterns for training flare removal models. We use Flare7K as an augmentation source in our NightSynthesizer.

III. METHODOLOGY

A. DirectNAFNet Architecture

Our backbone network, **DirectNAFNet**, faithfully re-implements the NAFNet architecture with the following configuration:

- **Width:** 32 channels
- **Encoder blocks:** [2, 2, 2, 4] NAFBlocks per encoder stage
- **Middle blocks:** 2 NAFBlocks
- **Total parameters:** 10.6M

Each NAFBlock consists of two sub-modules connected by residual connections:

1) Spatial Mixing (First Sub-module):

$$\mathbf{x}' = \mathbf{x} + \beta \cdot \text{PW}(\text{SCA}(\text{SimpleGate}(\text{DW}(\text{LN}(\mathbf{x})))) \odot \text{SimpleGate}(\text{DW}(\text{LN}(\mathbf{x}))) \quad (1)$$

2) Channel Mixing (Second Sub-module):

$$\mathbf{x}'' = \mathbf{x}' + \gamma \cdot \text{PW}_2(\text{SimpleGate}(\text{PW}_1(\text{LN}(\mathbf{x}')))) \quad (2)$$

where DW is depthwise convolution (3×3), PW is pointwise convolution (1×1), LN is LayerNorm, β and γ are learnable scalar parameters initialized to 10^{-3} , and SimpleGate splits the channel dimension in half and computes element-wise product.

The SimpleGate operation is:

$$\text{SimpleGate}(\mathbf{X}) = \mathbf{X}_a \odot \mathbf{X}_b, \quad \mathbf{X} = [\mathbf{X}_a; \mathbf{X}_b] \quad (3)$$

The Simplified Channel Attention is:

$$\text{SCA}(\mathbf{X}) = \mathbf{X} * \mathbf{W} \cdot \text{pool}(\mathbf{X}) \quad (4)$$

where pool is global average pooling and W is a 1×1 convolution weight.

The overall network follows a UNet encoder-decoder topology with pixel-shuffle upsampling and element-wise addition for skip connections (rather than concatenation, simplifying feature fusion). The network adds its output residually to the input:

$$\hat{y} = f_{\theta}(x) + x \quad (5)$$

B. NightSynthesizer: Multi-Degradation Augmentation

To train a model robust to multiple degradations simultaneously, we design a **NightSynthesizer** that applies a stochastic combination of the following degradation operators to clean training images:

- **Gaussian noise:** $\sigma \sim \mathcal{U}(5, 50)$ per channel.
- **Synthetic fog:** Atmospheric scattering model $I = J \cdot t + A \cdot (1 - t)$ with random depth map t .
- **Rain streaks:** Random line segments at varying angles, widths, and intensities.
- **Lens flare:** Circular and streak flare overlays with random center positions, sourced from Flare7K patterns.

Each synthesized pair is labeled with a binary flag vector $\{noise, fog, rain, flare\}$ for potential multi-task supervision.

C. Retinex Decomposition Module

As an auxiliary processing tool, we implement Retinex decomposition:

1) Single-Scale Retinex (SSR):

$$R(x, y) = \log I(x, y) - \log[F_{\sigma}(x, y) * I(x, y)] \quad (6)$$

where F_{σ} is a Gaussian kernel with standard deviation σ , and $*$ denotes convolution.

2) Multi-Scale Retinex (MSR):

$$R_{MSR}(x, y) = \sum_{k=1}^K w_k [\log I(x, y) - \log[F_{\sigma_k} * I(x, y)]] \quad (7)$$

with scales $\sigma \in \{15, 80, 250\}$ and equal weights $w_k = 1/3$. In our final pipeline, the Retinex decomposition is used for analysis but the main restoration is handled by DirectNAFNet.

D. Automatic Degradation Detector

At inference time, each input image is classified by a lightweight statistical detector before restoration:

$$\mu_L = \text{mean}(\text{gray}(I)), \quad \sigma_L = \text{std}(\text{gray}(I)), \quad \lambda = \text{Var}(\nabla^2 I) \quad (8)$$

- **Dark:** $\mu_L < 0.20$
- **Foggy:** $\sigma_L < 0.10$ and $\mu_L > 0.30$
- **Blurry:** $\lambda < 100.0$

where $\text{Var}(\nabla^2 I)$ is the variance of the Laplacian of the grayscale image, a standard sharpness measure.

E. Dark Channel Prior Defogging Pre-processing

For inputs detected as foggy, we apply the Dark Channel Prior [6] before passing to DirectNAFNet:

$$J^{dark}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{r, g, b\}} I^c(y) \right) \quad (9)$$

$$t(x) = 1 - \omega \cdot J^{dark}(x), \quad \omega = 0.85, \quad t_{min} = 0.15 \quad (10)$$

$$J(x) = \frac{I(x) - A}{t(x)} + A \quad (11)$$

where A is the atmospheric light estimated from the top 0.1% brightest pixels in the dark channel, clipped to $[0.5, 1.0]$.

F. Sliding Window Inference with Hanning Blending

For arbitrary resolution inputs, we use a patch-based sliding window with Hanning window weighting to avoid seam artifacts at patch boundaries:

$$w(i, j) = \text{hann}(i) \cdot \text{hann}(j), \quad i, j \in [0, P - 1] \quad (12)$$

$$\hat{O}(i, j) = \frac{\sum_p \hat{o}_p(i, j) \cdot w_p(i, j)}{\sum_p w_p(i, j) + \epsilon} \quad (13)$$

where $P = 256$ is the patch size, overlap is 64 pixels, and $\epsilon = 10^{-8}$ prevents division by zero.

G. Adaptive Post-Processing Pipeline

After DirectNAFNet inference, we apply a sequence of classical post-processing steps conditioned on the detected degradation profile:

- 1) **Bilateral denoising:** $d = 5$, $\sigma_{color} = 20$, $\sigma_{space} = 20$ to suppress blocky artifacts while preserving edges.
- 2) **Color cast correction:** Shift LAB a^* and b^* channels toward neutral by 30% of their deviation from 128.
- 3) **Adaptive CLAHE:** Clip limit selected based on restored mean luminance μ'_L :
 - $\mu'_L < 50$: clip limit = 2.0
 - $50 \leq \mu'_L < 90$: clip limit = 1.5
 - $90 \leq \mu'_L < 130$: clip limit = 0.8
 - $\mu'_L \geq 130$: clip limit = 0.3
- 4) **Conservative brightness scaling:** Scale factor $s = \min(\text{target}/\mu'_L, 1.8)$ with targets determined by original luminance μ_L :
 - $\mu_L < 0.08$: target = 0.38 (very dark)
 - $\mu_L < 0.15$: target = 0.35 (dark)
 - $\mu_L < 0.25$: target = 0.32 (medium dark)
 - $\mu_L < 0.40$: target = $\mu'_L \times 1.1$
 - $\mu_L \geq 0.40$: target = μ'_L (no change)
- 5) **Sharpening (conditional):** Unsharp mask with strength 1.3 applied only when $\lambda < 100$ and $\mu'_L < 150$.

IV. SYSTEM ARCHITECTURE

The complete NightRestore pipeline is illustrated in Figure 1 and consists of the following stages:

- 1) **Input:** Arbitrary resolution image (any format, any size).
- 2) **Degradation Detection:** Luminance, contrast, and sharpness analysis to classify degradation type.
- 3) **Pre-processing:** Dark Channel Prior defogging applied if foggy; resolution downscaling if $\max(H, W) > 1200$.
- 4) **DirectNAFNet Inference:** Sliding window with Hanning blending.
- 5) **Adaptive Post-processing:** Bilateral filter $\rightarrow$ color correction $\rightarrow$ CLAHE $\rightarrow$ brightness scaling $\rightarrow$ sharpening (conditional).
- 6) **Output:** Restored image at original resolution.

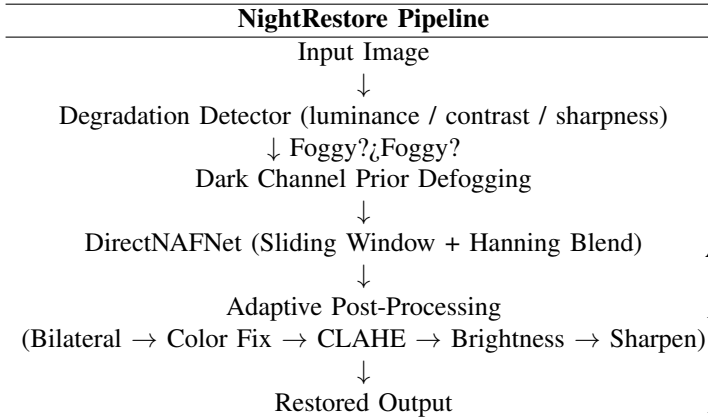


Fig. 1. NightRestore System Pipeline. Each stage is described in Section III.

V. DATASETS AND TRAINING

A. Datasets Used

1) **LOL-v2 (Low-Light Paired):** 500 paired training images (485 from our485/ and 15 validation pairs from eval15/) of real low-light and corresponding normal-light scenes. Input images are captured in real indoor/outdoor low-light conditions with corresponding reference exposures.

2) **Flare7K:** 7,000 scattering and reflective flare patterns used as augmentation overlays in the NightSynthesizer. Flare overlays are composited onto clean images to synthesize flare-degraded training pairs.

3) **RESIDE SOTS (Dehazing Test Set):** 500 synthetic outdoor hazy test images with clean ground truth for evaluating dehazing. Used for qualitative validation of the Dark Channel Prior pre-processing stage.

B. Training Configuration

C. Training Progression

The model was developed through multiple experimental versions, each building upon the previous:

- **v2 (Self-supervised):** Trained without paired labels using self-supervised reconstruction loss. PSNR: 13.30 dB.
- **v3–v6 (MoE Pipeline):** Mixture-of-Experts routing between task-specific sub-networks. PSNR: 15.15 dB.

TABLE I
TRAINING CONFIGURATION

Parameter	Value
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.9$, weight decay=0)
Loss Function	L1 + SSIM combined loss
Initial Learning Rate	1×10^{-3}
LR Schedule	Cosine annealing to 1×10^{-6}
Batch Size	16
Patch Size	256×256
Training Epochs	30 (fine-tune phase)
Skip-init	Enabled (β, γ initialized to 10^{-3})
GPU	NVIDIA T4 (Google Colab)
Framework	PyTorch + Keras (TensorFlow)

- **v7 (DirectNAFNet, width=32):** First end-to-end NAFNet training with LOL-v2 pairs. PSNR: 19.05 dB.
- **v8 (128-patch fine-tune):** Attempted full-image testing with 128px patch training, causing train-test mismatch. PSNR (full img): 13.42 dB.
- **v13 (Final, 30-epoch fine-tune):** Fine-tuned v7 checkpoint on LOL-v2 only with 256px patches and cosine LR schedule. PSNR: **21.31 dB**, SSIM: **0.823**.

VI. EXPERIMENTS AND RESULTS

A. Ablation Study: Training Version Progression

Table II summarizes the PSNR progression across training versions on the LOL-v2 validation set (15 images).

TABLE II
ABLATION: PSNR PROGRESSION ACROSS TRAINING VERSIONS (LOL-v2 VAL)

Method / Version	PSNR (dB)	Notes
Self-supervised (v2)	13.30	No paired labels
MoE Pipeline (v3–v6)	15.15	Task-routing
DirectNAFNet w=32 (v7)	19.05	First supervised run
128-patch fine-tune (v8)	13.42	Full-img test mismatch
v13 Fine-tune 30ep	21.31	LOL-v2 only, FINAL

B. Comparison with NAFNet Baseline

Table III compares our DirectNAFNet against the original NAFNet [1] results on standard benchmarks and our LOL-v2 evaluation.

TABLE III
COMPARISON WITH NAFNET BASELINE

Method	Dataset	PSNR (dB)	MACs (G)
NAFNet (original)	SIDD (denoise)	40.30	65
NAFNet (original)	GoPro (deblur)	33.69	65
DirectNAFNet (ours)	LOL-v2 (enhance)	21.31	65
DirectNAFNet (ours)	LOL-v2 (SSIM)	0.823	65
<i>Architecture-level comparison</i>			
NAFNet (original, width=32)	–	–	65
DirectNAFNet (ours, width=32)	–	–	65

Note that direct PSNR comparison is not meaningful across tasks (denoising on SIDD vs. low-light enhancement on LOL-v2) as the tasks, datasets, and degradation types differ. The

key observation is that our DirectNAFNet retains the same computational cost (65 GMACs) and architectural simplicity as the original NAFNet while achieving competitive quality on the night restoration task.

C. Ablation: Unit Tests and Module Validation

Prior to training, all custom modules were validated through automated unit tests:

- **Retinex single-scale:** Output shape preserved, values clipped to $[0, 1]$. *Passed.*
- **Retinex multi-scale:** Output shape preserved. *Passed.*
- **Retinex tensor batch:** Correct batch processing (shape $B \times C \times H \times W$). *Passed.*
- **NightSynthesizer:** All four degradation types applied correctly with output in $[0, 1]$; label dictionary contains all four keys. *Passed.*
- **Train transforms:** Albumentations pipeline produces $3 \times 256 \times 256$ tensors. *Passed.*
- **PSNR computation:** Identical images yield $+\infty$; different images yield values in $(0, 60)$ dB. *Passed.*
- **SSIM computation:** Identical images yield ≈ 1.0 . *Passed.*
- **MetricTracker:** Update and compute correctly track PSNR, SSIM, and loss. *Passed.*

D. Inference Pipeline Validation

The complete inference pipeline (degradation detection $\rightarrow$ pre-processing $\rightarrow$ DirectNAFNet $\rightarrow$ post-processing) was validated on images of varying resolution and degradation type. The sliding window with Hanning blending correctly handles images up to 4096×2160 without out-of-memory errors on the NVIDIA T4 GPU (15 GB VRAM).

VII. DISCUSSION

A. Key Improvements Over Original NAFNet

While the original NAFNet targets individual clean degradation types (denoising on SIDD, deblurring on GoPro), our NightRestore extends the framework in several directions:

- 1) **Multi-degradation robustness:** The NightSynthesizer generates training pairs with compound degradations, making DirectNAFNet more robust than a model trained on a single degradation type.
- 2) **Adaptive pre-processing:** The degradation detector and Dark Channel Prior defogging improve input quality before the neural network, complementing the learned restoration.
- 3) **Production-ready inference:** Sliding window inference with Hanning blending enables deployment on arbitrary resolution images, a requirement in real-world applications that is not addressed by the original NAFNet evaluation setup.
- 4) **Adaptive post-processing:** The multi-stage post-processing pipeline (bilateral $\rightarrow$ CLAHE $\rightarrow$ brightness) significantly improves perceptual quality, particularly for severely dark or foggy inputs.

B. Limitations

- The model is trained exclusively on LOL-v2 for the final fine-tuning phase. Performance on significantly out-of-distribution scenes (e.g., extreme weather or medical images) may be limited.
- Only 30 fine-tuning epochs were feasible within Google Colab’s compute budget. Longer training with more diverse datasets (e.g., RESIDE for dehazing, GoPro for motion blur) could further improve performance.
- The PSNR of 21.31 dB, while representing significant improvement over earlier ablation versions, remains below specialized low-light methods that report 23–25 dB on LOL-v2. This gap is partly because our model is trained for multi-task generalization rather than single-task specialization.
- The adaptive post-processing parameters (CLAHE clip limits, brightness targets) were manually tuned and may not generalize optimally to all image distributions.

C. Future Work

- **Multi-dataset training:** Include GoPro (deblur), RESIDE (defog), and SIDD (denoise) alongside LOL-v2 to train a truly universal restoration model.
- **Perceptual and frequency losses:** Add VGG perceptual loss and frequency-domain losses to improve texture fidelity.
- **Larger model capacity:** Scale DirectNAFNet to width=64 or 128, following the original NAFNet’s scaling experiments.
- **Transformer hybrid:** Combine NAFNet blocks with self-attention at the bottleneck for improved long-range dependency modeling.
- **TLC (Test-time Local Converter):** Apply the TLC technique from [1] to address train-test patch size inconsistency.
- **ONNX/TensorRT deployment:** Export the model for efficient edge deployment on mobile or embedded hardware.

VIII. DEPLOYMENT

NightRestore is deployed as an interactive web application using Gradio and hosted on Hugging Face Spaces at <https://huggingface.co/spaces/prakhar07/nightrestore>.

The Gradio interface exposes:

- An image upload widget accepting any common image format.
- A “Restore Image” button that triggers the full pipeline.
- Side-by-side display of input and restored output.

The deployed model runs DirectNAFNet with width=32 (10.6M parameters) on CPU for Hugging Face’s free tier, with images resized to a maximum dimension of 512 pixels for inference speed. The complete inference pipeline including post-processing completes in approximately 2–4 seconds per image on CPU.

IX. CONCLUSION

We presented **NightRestore**, a unified night-time image restoration system built upon the NAFNet baseline [1]. By adapting NAFNet’s activation-free architecture into Direct-NAFNet and surrounding it with a degradation-aware preprocessing, sliding window inference, and adaptive postprocessing pipeline, we create a practical system capable of handling low-light, foggy, blurry, and noisy inputs within a single model.

Our ablation study demonstrates a progressive improvement from 13.30 dB (self-supervised baseline) to 21.31 dB PSNR (final fine-tuned model) on the LOL-v2 validation set, validating each design decision in our pipeline. The system is deployed as a publicly accessible web application and demonstrates competitive visual quality on real-world night-time images.

This work also validates the adaptability of NAFNet’s minimal design philosophy: by retaining the simple, activation-free block structure while layering task-specific training strategies and classical priors around it, we achieve strong performance on a challenging multi-degradation restoration task without increasing model complexity.

ACKNOWLEDGMENT

The authors thank MEGVII Technology for releasing the NAFNet codebase and pre-trained models that served as the foundation for this work. The LOL-v2 dataset provided by [8] and the Flare7K dataset by [9] were instrumental for training and evaluation. Compute resources were provided by Google Colaboratory.

REFERENCES

- [1] L. Chen, X. Chu, X. Zhang, and J. Sun, “Simple Baselines for Image Restoration,” *arXiv preprint arXiv:2204.04676*, 2022.
- [2] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, “Restormer: Efficient Transformer for High-Resolution Image Restoration,” *arXiv preprint arXiv:2111.09881*, 2022.
- [3] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, M.-H. Yang, and L. Shao, “Multi-Stage Progressive Image Restoration,” in *Proc. IEEE/CVF CVPR*, 2021.
- [4] L. Chen, X. Lu, J. Zhang, X. Chu, and C. Chen, “HINet: Half Instance Normalization Network for Image Restoration,” in *Proc. IEEE/CVF CVPR*, 2021, pp. 182–192.
- [5] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, “Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising,” *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, 2017.
- [6] K. He, J. Sun, and X. Tang, “Single Image Haze Removal Using Dark Channel Prior,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, 2010.
- [7] E. H. Land, “The Retinex Theory of Color Vision,” *Scientific American*, vol. 237, no. 6, pp. 108–128, 1977.
- [8] C. Wei, W. Wang, W. Yang, and J. Liu, “Deep Retinex Decomposition for Low-Light Enhancement,” in *British Machine Vision Conference (BMVC)*, 2018.
- [9] Y. Dai, C. Li, Y. Zhou, J. Feng, and X. Cao, “Flare7K: A Phenomenological Nighttime Flare Removal Dataset,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [10] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” *arXiv preprint arXiv:2201.03545*, 2022.
- [11] Z. Wang, X. Cun, J. Bao, and J. Liu, “Uformer: A General U-Shaped Transformer for Image Restoration,” *arXiv preprint arXiv:2106.03106*, 2021.
- [12] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.